



UNIVERSITY OF
WATERLOO

MTE 203 – Lecture 9

Midterm Exam Fall 2016 – Problem 2

Midterm Exam Fall 2016 – Problem 2

A curve of intersection is defined by the surfaces given below, for the values in the positive octant only.

$$x^2 + y^2 + z^2 = 4, \quad x + z = 2$$

- a. Sketch the curve of intersection
- b. Find a vector valued function for the curve directed so that z decreases from 2 to 0 using the parametrization,

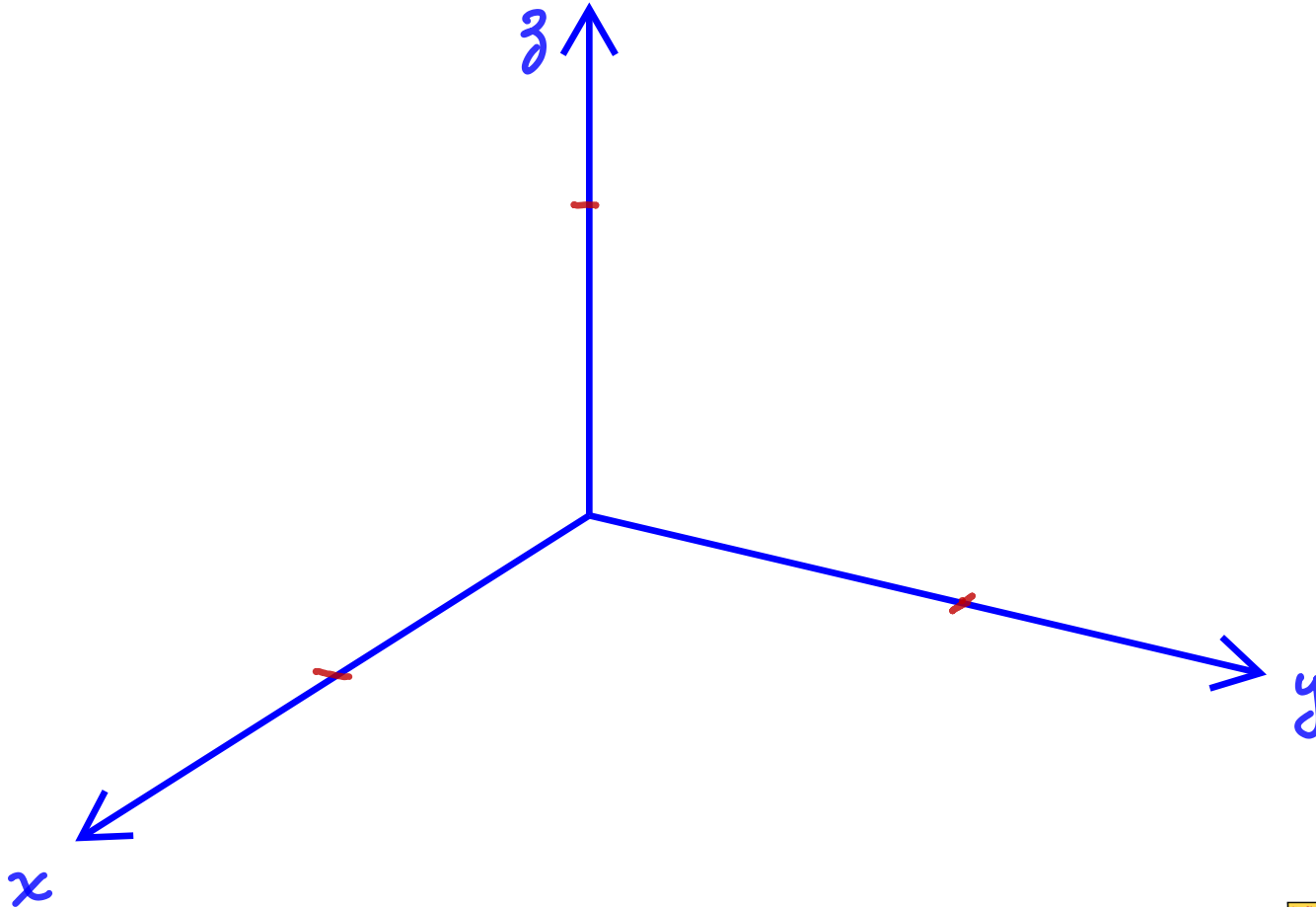
$$x = 1 + \sin t$$

- c. What is the range of the parameter t for the curve of intersection? Clearly indicate the direction of the curve in your sketch.
- d. Calculate the unit tangent vector $\hat{T}$ in the direction of increasing t .
- e. Find the length of the curve obtained in parts b and c
- f. Without calculating the moving right-handed Frenet vector Frame, can you determine the equation of the osculating plane at the point $(1, \sqrt{2}, 1)$ in the curve?



Midterm Exam Fall 2016 – Problem 2

a. Sketch of the curve of intersection



Midterm Exam Fall 2016 – Problem 2

- b. Find a vector valued function for the curve directed so that z decreases from 2 to 0 using the parametrization,

$$x = 1 + \sin t$$



Midterm Exam Fall 2016 – Problem 2

- c. What is the range of the parameter t for the curve of intersection? Clearly indicate the direction of the curve in your sketch.



Midterm Exam Fall 2016 – Problem 2

d. Calculate the unit tangent vector $\hat{T}$ in the direction of increasing t .



Midterm Exam Fall 2016 – Problem 2

- e. Find the length of the curve obtained in parts b and c



Midterm Exam Fall 2016 – Problem 2

- f. Without calculating the moving right-handed Frenet vector Frame, can you determine the equation of the osculating plane at the point $(1, \sqrt{2}, 1)$ in the curve?

